



Research article

The association between exposure to climate change events and aggression among university students: A cross-sectional study

Muna Ahmead^{a,*}, Nuha El Sharif^a, Etaf Maqboul^b, Raja Zyoud^c, Inad Nawajah^d^a Faculty of Public Health, Al-Quds University, Jerusalem P.O. Box 51000, Palestine^b Faculty of Nursing & Health Sciences, Bethlehem University, Palestine^c Faculty of Nursing, The Arab American University, Palestine^d Department of Mathematics, College of Science & Technology, Hebron University, Palestine

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ABSTRACT

Background: Climate change has been linked to increased psychological distress, aggression, and violence. This study aimed to assess the association between reported climate-related event exposure and aggressiveness among university students.**Methods:** The study, a cross-sectional research design, utilized the Climate Change Anxiety Scale, Brief Aggression Questionnaire, and Kessler Psychological Distress Scale. The relationship between scores and reported climate-related event exposure was investigated using frequencies, percentages, and multivariate analysis.**Results:** 1338 students were recruited. Survey results showed that 23.5% of students who reported climate-related event exposure exhibited aggression, 35.3% were angry, 27.7% were verbally aggressive, 30.1% were hostile, and 24.3% were physically aggressive. Multivariate analysis showed that the probability of reporting exposure to a climate-related event was higher among females, those with low academic performance (AOR: 2.22, p-value < 0.001), and those with a negative mood (AOR: 2.94, p-value < 0.001). Cognitive-emotional climate anxiety was associated with decreased reporting of climate-related event exposure (AOR: 0.37, p-value = 0.005). Participants with scores in the range of increased risk for serious mental illnesses were twice as likely to report a climate-related event exposure (AOR: 2.13, p-value < 0.001). Moderate and high anger were associated with decreased reporting of exposure to climate-related events.**Conclusion:** Climate-related event exposure is associated with increased aggression among undergraduate students. Furthermore, climate-related event exposure is more frequently reported among women, those with lower academic achievement, negative moods, and potential serious mental illnesses. Participants with moderate to high anger and cognitive-emotional climate anxiety are less likely to report climate-related event exposure. To reduce aggression associated with climate event exposure, universities should consider providing psychological and counseling services, as well as climate-related event exposure and stress adaptation strategies.

1. Introduction

Climate change is a serious public health problem in the twenty-first century [1]. More frequent and catastrophic heat waves and floods are putting a strain on people's mental health [2]. It has been associated with psychological distress, sadness, loss, and discontent [3,4]. According to Ojala et al [5], exposure to discussions about climate change, potential future consequences, uncertainties, and the continent's sufferings may also affect mental and emotional health. Climate change and inaction may put young adults worldwide at risk for mental health problems [6–8]. In a study of young British adults, 74% were concerned

about how climate change might influence their future and 63% about its consequences [9]. Mental health practitioners have discussed "eco-anxiety" and "climate anxiety" in relation to climate change awareness [10], particularly among young people [6,11,12]. "Eco-anxiety" is a new term for climate-related anxiety and emotional reactions [13]. People experience climate change anxiety (CCA) when their natural environment changes and becomes uncertain [10]. It influences cognitive, emotional, decision-making, and behavioral responses [11,14,15]. CCA can induce fear, anxiety, anger, despair, guilt, frustration, and grief [13,16,17]. The global climate crisis disproportionately impacts low-income populations, children, youth, the elderly, and the

* Corresponding author.

E-mail address: munaahmead@yahoo.com (M. Ahmead).<https://doi.org/10.1016/j.joclim.2025.100640>

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disabled [18]. A global survey reveals that 45% of 16–25-year-olds believe climate change is adversely impacting their daily lives and performance [6].

The weather and climate, two related yet distinct concepts, profoundly influence our natural environment and several aspects of human life. The transient atmospheric conditions in a certain region, encompassing elements such as temperature, humidity, precipitation, wind velocity, and air pressure, are termed "weather," which is the short-term, day-to-day or week to week atmospheric conditions. "Climate" refers to the long-term average of numerous meteorological elements over extended periods, typically spanning decades to centuries, while intermittent events, such as heat waves, are classified as severe weather events [19,20].

Many research studies have investigated the impact of meteorologic variables on mental health, but few have examined climate change exposure and aggression [21]. The term "aggression" refers to any behavior intended to damage or injure another individual [22–24]. Aggressive behavior may be classified as physical, verbal, and relational aggression, along with two functions: proactive/instrumental and reactive/defensive aggression. Aggression also may be classified as reactive, instrumental (proactive), or appetitive type [24–26]. Numerous psychological theories elucidate aggression. A prominent framework is the social learning hypothesis, which asserts that aggressive behavior is acquired through observation and reinforcement in one's surroundings. The frustration-aggression hypothesis asserts that aggressiveness directly arises from frustration, especially when individuals perceive obstacles to their goal attainment [27]. Also, cognitive-behavioral perspectives assert that maladaptive thoughts may enhance aggressiveness [28].

The association between exposure to climate change and aggression has been the subject of some research. One study found a correlation between greater yearly temperatures and more violent crime, such as homicide and assault [29]. Sanz-Barbero et al. found that heatwaves in Spain increased intimate partner femicides [30]. Mares and Moffetti studied 57 countries from 1995 to 2012 and estimated that crime rates to rise by 6% for every degree Celsius that climate change raises the world average temperature [21]. A meta-analysis found a correlation between climate and aggressiveness in 46 of 56 studies (82%) [31]. Burke et al. (2015) found a connection between temperature and violent behavior in 20 of the 24 studies (83%). However, other research revealed no relationship between climate change and crime [32].

In addition to associations between climate change event exposure and aggression, feelings of climate anxiety can be connected to many emotions, including worry, fear, anger, grief, despair, guilt, and shame, as well as hope [6]. Emotional responses to climate change demonstrate that individuals are concerned, and negative feelings reflect dissatisfaction with the present circumstances [33]. While emotions like fear and hope have garnered significant investigation, anger is among the least examined emotional responses to climate change, even though it is among the most frequently associated emotions [34–37]. The assessment of eco-anger generally involves asking if individuals experience anger over climate change or requests that they evaluate their anger on a continuous scale [38]. For instance, Marczak et al. established the Inventory of Climate Emotions (ICE) to provide a comprehensive and integrated evaluation of eco-emotions including anger, contempt, enthusiasm, powerlessness, guilt, isolation, and anxiety [36,39]. The Climate Change Distress Scale (CCDS) is a 21-item instrument intended to evaluate emotional distress associated with climate change, including emotions like anger, rage, and frustration. Although it lacks a specific subscale for anger, it assesses total distress, which frequently has anger as a significant component [40]. Hickman et al. conducted a study involving 10,000 children and young adults aged 16–25 years from 10 countries examining climate-related emotions. Over 50% of respondents reported experiencing 14 important positive and negative emotions regarding climate change, including feeling afraid, sad, anxious, angry, powerless, helpless, and guilty [6]. In the study by Gregersen et al.,

climate anger was assessed using the question, "When it comes to climate change and everything you associate with it, how strongly do you experience the following emotions?" Anger was listed together with sadness, fear, guilt, and hope. The study revealed that individuals reporting a higher level of anger expressed more robust intentions to participate in activism, policy support, and individual pro-environmental behavior [41].

Aggression in general is common among university students [42], ranging from 35.4% to 51.3% in different studies [42–44] but little is known about its relation to exposure to climate change-related events. Most climate change-related mental health research focuses on depression, anxiety, and PTSD, and most has been based in Western countries [45–48,21].

Palestine's eastern Mediterranean location between the Sinai Peninsula and the Jordanian desert influences its climate. During the three months of winter in Palestine, temperatures can drop below freezing. The hottest months in Palestine are July and August, and temperatures can sometimes reach minimum 30 °C (86°F) [49]. Reflecting long-term changes, droughts have increased, precipitation has decreased, average temperatures have risen, rainfall has declined, and winter temperatures have dropped; the number of unusually hot summer days has increased, and both minimum and maximum summer temperatures have risen [50]. Palestine lacks research on climate change event exposure-related mental health problems among young people [51]. We hypothesized that there is a positive association between exposure to climate-related events and aggression among university students; thus, the objective of this study was to assess this association. The study also examined the association between climate-related event exposure and academic performance, self-control, temperature, climate anxiety, and psychological distress. Finally, it examined the predictors of climate-related event exposure among Palestinian undergraduate students.

2. Methods

2.1. Study design and sampling

The study utilized a cross-sectional study design, and was undertaken between September 1 and October 30, 2024. It included undergraduates in the West Bank from 14 public universities. Participants were recruited using convenience sampling, a non-probability method that selects participants from a population of interest based on their availability, due to the Israeli military's blockade of Palestinian cities and mobility restrictions in the West Bank, which challenged the recruitment of distance learning students. A power analysis established a minimum sample size of 1,015 students with a significance level of 0.05, a confidence level of 95%, and an accuracy of 0.02. We anticipated a 30% non-response rate. The ultimate sample size calculation was 1,320 participants.

2.2. Tools and measures

A self-administered anonymous online survey was utilized for data collection. We sent an introduction and invitation to participate via an online Google Form questionnaire to student groups on Facebook, social media platforms, emails, and WhatsApp. This questionnaire included the following:

- Socio-demographic data included age, gender, marital status, family monthly income, education level, faculty (department/school), year of study, district, specialty, and place of residence. Participants were asked whether they have been exposed to climate change-related events such as severe temperature fluctuations, droughts, rising sea levels, storms, floods, wildfires, or precipitation changes. Their response was categorized as "yes" or "no". If they replied affirmatively, we asked them to note the primary type of climate change

event that affected them. Their responses were categorized into three groups: extreme temperature change (increase or decrease in temperature), droughts, and floods. Furthermore, they were asked if their study, mood, and self-control of their behavior were negatively affected by climatic changes as follows: Does climate change have a negative effect on your academic achievement? Does climate change have a negative effect on your mood? And have you experienced loss of control of your behavior because of climate change?

- The Climate Change Anxiety Scale [11]; this was used to assess climate anxiety. It is composed of 13 items rated on a five-point Likert-type scale from 1 = "never" to 5 = "almost always," with two domains measured including cognitive-emotional (items 1–8) and functional impairment (items 9–13) due to climate change. Cut-off scores over the midpoint (3) were considered a sign of climate anxiety, according to the criteria of Clayton and Karazsia [11]. The internal consistency coefficient (Cronbach's α) was 0.94.
- The Kessler Psychological Distress Scale-6 (K6) [52]; items assess feelings of nervousness, hopelessness, restlessness, jumpiness, sadness, and worthlessness. Each item of the K6 is assessed using a 5-point Likert-type intensity scale: None of the time (0); a little of the time (1); some of the time (2); most of the time (3); and all of the time (4). The K6 items are rated from 0 to 4, with the final score being the aggregate of these responses, ranging from 0 to 24. A score of 13 or above suggests a likelihood of serious mental illness. It measures the following six symptoms: felt nervous, hopeless, restless or fidgety, worthless, depressed, and felt that everything was an effort. A score under 13 suggests that a severe mental illness is unlikely. The internal consistency coefficient (Cronbach's α) was 0.83.
- Brief Aggression Questionnaire (BAQ); this is a 12-item self-report measure of the trait of aggression [53]. The questionnaire asks participants to rate on a scale from 1 (strongly agree) to 5 (strongly disagree) the degree to which statements describing behaviors and emotions are characteristic of themselves. The BAQ measures aggression in the domains of physical aggression, verbal aggression, anger, and hostility. The internal consistency coefficient (Cronbach's α) was 0.97.

A committee of four mental health experts reviewed the study questionnaire's contents including all scales to ensure that the tool was culturally appropriate, and no changes were made. The scale was first translated into Arabic by the research team and then back-translated to English by a licensed translator. At the pilot stage, we administered the tool to 25 undergraduates to test for language clarity; both the original English questionnaire and the back-translated version were examined to ensure that the translation was accurate.

2.3. Data analysis

The data were analyzed with SPSS version 25 (IBM Corp., Chicago, IL, United States). The descriptive analysis for all study variables is reported as frequencies and percentages. The chi-square test was used to investigate the associations between climate change exposure and study variables. A multivariate regression analysis was carried out, and the results are presented as adjusted odd ratios (AOR) with a 95% confidence range with a p-value of less than 0.05 considered significant.

2.4. Ethical consideration

This study adhered to the Declaration of Helsinki in all its methods. The study was approved by the Al Quds University Research Ethics Committee (Ref No: 404/REC/2024). This online survey was anonymous. Written information was provided at the beginning of the study about the aim of the study and how the data would be used. Upon filling out the questionnaire, students provided informed consent for participation in this study.

3. Results

3.1. Socio-demographic characteristics of the study population

1,338 students completed the questionnaire. Table 1 shows that the majority of participants were females (69.4%) residing in urban areas (47.4%); 88.3% were unmarried, 47.2% were enrolled in health science faculties/departments and 50.7% were living in the southern region of the West Bank. Table 1 shows that 66% of the participants reported exposure to climate-related events, of which 80% of them reported experiencing extreme temperature change (increase or decrease in temperature), 15% experienced droughts, and 5% experienced floods.

3.2. Association between climate-related event exposure and study variables

As shown in Table 2, there were significant associations between climate-related event exposure and self-reported academic achievement, moods affected negatively by climate change, loss of self-control of behavior, distress, aggression, anger, verbal aggression, hostility (with a p-value < 0.001), and physical aggression (with a p-value of 0.021). Also, the findings revealed that among the participants who

Table 1
Socio-demographic characteristics of the study population and climate-related event exposure (N = 1338).

		F	%
Gender	Male	409	30.6 %
	Female	929	69.4 %
Place of residency	City	634	47.4 %
	Village	605	45.2 %
Family income per month (US dollar)	Refugee camp	99	7.4 %
	No income	174	13.0 %
	<270	82	6.1 %
	270–540	288	21.5 %
	541–1080	391	29.2 %
	1081–1620	213	15.9 %
	>1620	190	14.2 %
Marital status	Single	1181	88.3 %
	Not single (divorced or widowed)	157	11.7 %
Faculties	Science faculties	381	28.5 %
	Faculties of art (Humanities)	326	24.4 %
Year of Study	Health sciences faculties	631	47.2 %
	1st	272	20.3 %
	2nd	276	20.6 %
	3rd	391	29.2 %
	4th	339	25.3 %
Governorate of living	5th – 6th	60	4.5 %
	North	348	26.0 %
	Middle	312	23.3 %
	South	678	50.7 %
Did you expose to climate-related events?	Yes	885	66 %
	No	453	40 %

Table 2

The association between exposure to climate change event and self-reported academic achievement, self-control, mood, temperature, climate anxiety, aggression, and psychological distress.

		Exposure to climate related event				
		No		Yes		
		N = 453		N = 885		
		N	N %	N	N %	P value
Did climate relate event exposure has a negative effect on your academic achievement?	Yes	184	40.6%	508	57.4%	<0.001
	No	269	59.4%	377	42.6%	
Did climate related event exposure has a negative effect on your mood?	Yes	300	66.2%	767	86.7%	<0.001
	No	153	33.8%	118	13.3%	
Did you lose self-control over behavior due to climate-related event exposure?	Yes	198	43.7%	528	59.7%	<0.001
	No	255	56.3%	357	40.3%	
Climate Anxiety Scale:						
Total Cognitive Climate Anxiety	No anxiety	428	94.5%	828	93.6%	0.51
	Sign of climate anxiety	25	5.5%	57	6.4%	
Cognitive- Emotional Climate Anxiety	No anxiety	429	94.7%	845	95.5%	0.52
	Sign of climate anxiety	24	5.3%	40	4.5%	
Functional Climate Anxiety	No anxiety	418	92.3%	788	89.0%	0.06
	Sign of climate anxiety	35	7.7%	97	11.0%	
Kessler Psychological Distress Scale	Unlikely	346	76.4%	465	52.5%	<0.001
	Serious mental illness	107	23.6%	420	47.5%	
Aggression Scale:						
Brief Aggression Scale	Low	82	18.1%	251	28.4%	<0.001
	Moderate	243	53.6%	426	48.1%	
	High	128	28.3%	208	23.5%	
Anger subscale	Low	112	24.7%	344	38.9%	<0.001
	Moderate	179	39.5%	229	25.9%	
	High	162	35.8%	312	35.3%	
Verbal subscale	Low	93	20.5%	271	30.6%	<0.001
	Moderate	260	57.4%	369	41.7%	
	High	100	22.1%	245	27.7%	
Hostility subscale	Low	94	20.8%	275	31.1%	<0.001
	Moderate	235	51.9%	344	38.9%	
	High	124	27.4%	266	30.1%	
Physical subscale	Low	210	46.4%	459	51.9%	.021 *
	Moderate	101	22.3%	211	23.8%	
	High	142	31.3%	215	24.3%	

* p < 0.05.

reported exposure to climate-related events, 23.5% reported a high level of aggression, 35.3% had a high level of anger, 27.7% had a high level of verbal aggression, 30.1% had a high level of hostility, and 24.3% had a high level of physical aggression.

3.3. Multivariate logistic regression for determinants of climate-related event exposure

In Table 3, the multivariate logistic regression analysis revealed that being male decreased the chance of reporting climate-related event exposure (AOR: 0.44, p-value < 0.001). The participants who reported that the climate affected their academic achievement were nearly twice as likely to report climate-related event exposures compared to those who did not report an exposure (AOR: 2.22, p-value < 0.001). Also, the participants who reported that their mood was negatively affected by

Table 3

Multivariate analysis for climate-related event exposure and aggressive behavior.

		Sig.	AOR	95% CI for AOR	
				Lower	Upper
Gender	Male/Female	<0.001	0.44	0.33	0.60
Did climate-relate event exposure has a negative effect on your academic achievement?	(Yes/No)	<0.001	2.22	1.66	2.95
Did climate-related event exposure has a negative effect on your mood?	(Yes/No)	<0.001	2.94	2.14	4.03
Cognitive- emotional Climate Anxiety (CCA)	Sign/no anxiety	0.005	0.37	0.18	0.74
Kessler Psychological Distress Scale (KPD) Score	Serious mental illness/Unlikely	<0.001	2.13	1.59	2.84
Anger subscale	Low	Reference			
	Moderate	0.014	0.47	0.26	0.86
	High	0.002	0.35	0.18	0.69
Verbal subscale	Low	Reference			
	Moderate	0.77	0.91	0.495	1.69
	High	0.11	1.85	0.86	4.01

AOR: Adjusted Odds Ratio, 95% CI: 95% Confidence Interval, Sig.: Significance level (p < 0.05). Odds ratios reflect the likelihood of subjects reporting climate related event exposure.

The multivariate logistic regression model controls for reported academic achievement, mood negativity, and self-control behavior. In addition, the Total Cognitive Climate Anxiety (CCA) Score, Cognitive- Emotional Climate Anxiety, Functional Climate Anxiety, Total KPD score, Aggression Brief Scale, Aggression physical Scale, Aggression Anger Scale, Aggression verbal, and Aggression hostility were included after controlling for the participants' social demographic variables (age, gender, place of residence, family monthly income, marital status, faculty, year of study, and district of living).

climate change were nearly two times more likely to report climate-related event exposures compared to those who did not (AOR: 2.94, p-value < 0.001). In addition, the participants who reported cognitive-emotional climate anxiety (had signs of climate anxiety) were less likely to report climate-related event exposures than the participants who had low climate anxiety scores (AOR: 0.37, p-value 0.005).

Participants who scored in the high risk range for mental illness on the K6 scale were twice as likely to report climate-related event exposures compared to the participants who scored in the "unlikely" range (AOR: 2.13, p-value < 0.001). With respect to aggression associated with climate related event exposure, those having high anger scores on the BAQ were less likely to report climate-related event exposures (AOR=0.35, CI: 0.18–0.69); similarly, those who had moderate anger were also less likely to report climate-related event exposures (0.47, 95% CI: 0.26–0.86) when compared to low anger aggression level. Verbal aggression increased with climate-related event exposure but this association was not significant.

4. Discussion

Our results showed that anger was the only type of aggressiveness that was negatively associated with climate-related event exposure. Participants with moderate to high levels of anger were less likely to report climate-related event exposure than those with low levels of anger. Wesselbaum's research also found a negative association between climate change and aggression [54]. Nonetheless, our findings regarding anger contradict other prior research suggesting a positive relationship between climate-related event exposure and anger [55–57]. Spaiser et al. reported that exposure to the threat of climate change increases individuals' anger, which increases their willingness to participate in measures to mitigate climate change [58]. Palestine's relatively mild climate in comparison with other tropical climates may contribute to a

negative correlation between anger and climate-related event exposure. According to Choi et al., regions with tropical and humid climates have a positive association between temperature and crime, violence, or both (e.g., criminal arrest, homicide, assault, robbery, burglary), whereas studies in temperate climate countries show positive, negative, or mixed relationships [55]. Therefore, further longitudinal studies are required to investigate the relationship between aggression and climate change in different regions and with different manifestations of climate change.

With respect to other forms of aggressiveness, a significant number of participants exposed to climate-related events reported higher levels of aggression, including both verbal and physical aggression, as well as increased hostility and anger. This may suggest a psychological impact. Few studies examine the prevalence of aggressiveness in connection to climate-related event exposure. The present finding is significant because it provides evidence of the effect of climate-related event exposure on behavior, specifically an increase in aggressiveness measured at the level of the individual, rather than the population. One experimental study that did test individuals found that after being randomly assigned to sit in an uncomfortably hot room rather than a room at a comfortable temperature, heat-exposed participants were more likely to be hostile, perceive others as hostile, and act aggressively toward another person [59]. At the population level, a research study including sixty different countries estimated that for every degree (Celsius) increase attributable to climate change, homicide rates will rise by 6% [21]. Another study revealed that the total probability of assault mortality rose by 1.4% for every 1 °C increase in ambient temperature, notably among men, youths, and individuals with less education [20].

The increase in aggressive behavior on college campuses has significant ramifications. It jeopardizes the safety and well-being of students and teachers while undermining the fundamental mission of higher education institutions. Universities are intended to serve as environments for education, development, and intellectual discourse. Nonetheless, when aggressiveness becomes widespread, it undermines the academic atmosphere, generating tensions and obstructing the unrestrained exchange of ideas [60].

Our findings serve as a warning to educational stakeholders and professionals, necessitating the examination of characteristics that predict aggressive behavior among college students to facilitate their mitigation. Consequently, it is essential to implement climate change educational programs and stress management initiatives to assist students in addressing their challenges and enhancing their mental well-being.

The study found several factors that increased climate-related event exposure. Aligning with previous studies [61–63], our results showed that being male decreased the chance of reporting climate-related event exposures. Climate change has exacerbated violence against women by affecting economies, food supply, public health, water, education, and displacements [64]. To achieve the Sustainable Development Goals (SDGs) of reducing poverty, improving health and education, creating more sustainable cities, and combating climate change, a comprehensive strategy to protect girls and women during natural disasters is required [64]. Furthermore, students who reported low academic achievement were nearly twice as likely to report climate-related event exposures, consistent with prior studies [65–69]. According to Charity et al., climate warming resulted in 46.77% of students skipping 3–5 classes and 32.26% missing twice [65]. To address the consequences of climate change on academic performance, universities must provide undergraduates with counseling and psychological interventions.

With respect to climate change exposure and mood, participants who claimed that climate change exposure had a negative effect on their mood were nearly twice as likely to report climate-related event exposure compared to those who responded negatively. Schwaab et al. found that 3% of university students experienced depression symptoms associated with climate change [70]. Ballew et al. found that exposure to climate change events was linked to 16% of individuals experiencing depression or anxiety [71]. Negative emotions, such as sadness, are

deactivating and unlikely to stimulate action, and eco-depression can impede climate action [72]. Therefore, students who encounter climate change events are more likely to experience negative emotions, potentially impacting their overall mental health.

Furthermore, our findings revealed that those experiencing cognitive-emotional climate anxiety were less likely to report climate-related event exposure compared to those without this anxiety. Climate anxiety may include cognitive, emotional, and behavioral responses associated with climate change [10,17,73]. Cognitive-emotional climate anxiety might indirectly reduce a person's overall level of concern by inspiring individuals to engage in pro-environmental behaviors. Individuals may adopt personal lifestyle modifications to mitigate climate change risks, such as utilizing public transportation, decreasing meat consumption, saving energy, or engaging in recycling initiatives [74]. Thus, a degree of fear and anxiety over climate change might be adaptive, as it may incentivize pro-environmental activities aimed at mitigating its impacts [10,17,73].

With respect to other aspects of psychological/psychiatric symptoms, participants who scored high on the scale detecting likelihood of serious mental illness were twice as likely to report climate-related event exposure compared to those whose scores indicated little likelihood of serious mental illness. Several studies have found a correlation between climate change and an increase in mental crises and psychiatric hospital admissions [75,76]. In the context of psychoses [77] and cognitive disorders such as dementia [78], there is a substantial positive relationship between high temperatures and hospital admissions.

Our study's findings of both negative and positive associations between climate anxiety and distress are not inherently contradictory; they indicate a complicated association. A key explanation is the distinction between adaptive and maladaptive coping mechanisms. For certain individuals, climate anxiety may serve as a motivator that prompts proactive, problem-oriented coping strategies such as activism, lifestyle modifications, or participation in community initiatives. This sense of action and efficacy might diminish emotions of helplessness and despair, resulting in a negative association between climate anxiety and distress [79]. Conversely, climate anxiety may become so intense that it induces thoughts of hopelessness or paralysis. Such thinking may lead to maladaptive coping mechanisms, such as avoidance or denial, which can intensify feelings of distress and foster positive associations between climate anxiety and distress [74]. Another possible explanation is that in Palestine, general distress may be predominantly influenced by immediate stressors such as occupation or insecurity and wars, perhaps eclipsing climate-related concerns and creating challenges to cope with climate change.

4.1. Limitations

The current study uses validated and commonly used clinical questionnaires to measure climate anxiety, aggressiveness, and perceived stress in a broad population. Convenience sampling may induce selection bias. However, convenience sampling may be appropriate in conflict zones owing to the difficulty of examining a large population, although stratified or randomized samples may be more robust. Cross-sectional research cannot establish a causal association between climate-related event exposure and psychological effects. Therefore, longitudinal data is essential. Additionally, the climate change questionnaire is broad and subjective, potentially leading to recall bias and diminishing precision. The research questions might be further customized to the particular context of the West Bank (e.g., floods that may be associated with infrastructure failures rather than exclusively climate-related events).

Finally, while the study concluded that climate-related event exposure has an association with aggression, baseline data on the aggression of Palestinian university students is lacking. More research is needed to identify additional variables that could potentially contribute to university student aggression. Despite these limitations, our study found

that climate-related event exposure may have a detrimental influence on student mental health.

4.2. Implications for practice

The current study's findings may show the benefit of undertaking immediate climate action to minimize aggression among young adults. It also underlines the need for assisting undergraduate students in coping with hazards and increasing resilience to the effects of climate change and reducing its effect on their behavior. Educational institutions might establish centers for the development, testing, and dissemination of knowledge about climate change mitigation and adaptation techniques to lessen its influence on students' moods and improve their academic performance. Furthermore, colleges should have sufficient therapy facilities to address mental health concerns related to climate-related event exposure and aggression, which would improve student mental health and well-being. Longitudinal and qualitative studies are required to better understand the nature, causes, and progression of aggression in relation to climate-related event exposure, for which the present study provides a foundation. Future longitudinal data is required to prove a real cause-and-effect link between climate-related event exposure and other mental health issues among university students, such as depression, anxiety, and post-traumatic stress disorder (PTSD).

5. Conclusion

Climate change may increase aggression among undergraduates. Climate-related event exposure especially affects women, and is associated with negative effects on academic achievement, moods and incidence of serious mental illnesses. Participants with moderate to high anger and cognitive-emotional climate anxiety are less likely to report climate-related event exposure. To reduce aggression associated with exposure to climate-related events, universities should provide psychological and counseling services, as well as climate-related event exposure and stress adaptation strategies.

Declaration

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by Al Quds University Research Ethical Committee (Ref No: 404/REC/2024). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

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Declaration of generative AI

The authors declare that they did not use AI for manuscript preparation.

CRedit authorship contribution statement

Muna Ahmead: Writing – review & editing, Writing – original draft,

Validation, Supervision, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Nuha El Sharif:** Writing – review & editing, Validation, Formal analysis. **Etaf Maqboul:** Writing – review & editing, Validation, Data curation. **Raja Zyoud:** Writing – review & editing, Validation, Data curation. **Inad Nawajah:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

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